

Generative AI Search Engines as Arbiters of Public Knowledge: An Audit of Bias and Authority

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ABSTRACT

This paper reports on an audit study of generative AI systems (ChatGPT, Bing Chat, and Perplexity), which investigates how these new search engines construct responses and establish authority for topics of public importance. We collected system responses using a set of 48 authentic queries for 4 topics over 7 days, and analyzed the data using sentiment analysis, inductive coding, and source classification. Results provide an overview of the nature of system responses across these systems, and provide evidence of sentiment bias based on the queries and topics, and commercial and geographic biases in sources. The quality of sources used to support claims is uneven, relying heavily on News and Media, Business and Digital Media websites. Implications for system users emphasize the need to critically examine Generative AI system outputs when making decisions related to public interest and personal well-being.

KEYWORDS

Generative AI; search engines; sentiment analysis; information quality; cognitive authority

INTRODUCTION

The introduction of Generative AI (GenAI) Internet search tools constitutes the biggest shift in public access to information since Google and other second-generation search engines became available over 25 years ago. Systems such as Google Gemini, Microsoft Copilot, and Perplexity, mimic the familiar interaction paradigm of traditional search engines: users submit a query (or prompt), and the system responds with information and links to Internet sources. Yet, these new systems function very differently by drawing upon large language models, such as ChatGPT, to generate narrative responses and engage in dialogue with information seekers, rather than directing them to relevant web pages. *Large language models* (LLMs) are systems trained on large datasets to predict the upcoming words and phrases based on preceding and surrounding context (Bender et al., 2021). By designing systems that can synthesise information and make claims in direct response to people's queries, GenAI search engines are being set up as powerful gatekeepers and authorities in the public knowledge sphere, with the potential to surpass even Google's enormous influence (Haider & Sundin, 2019; Mager et al., 2023). Nevertheless, these systems are developing so quickly that we know very little about how they work and their likely impacts on individuals and society.

There are many reasons to be wary of this situation. LLMs are extremely powerful but prone to errors: generation of convincing misinformation (hallucination) (Bubeck et al., 2023; Ji et al., 2023; White, 2024), perpetuation and amplification of biases acquired through the model training process (Bender et al., 2021; White, 2024), dissemination of disinformation (Shah & Bender, 2022), and limited or incorrect attribution of sources (Kelly et al., 2023; Li et al., 2023; Liu et al., 2023; Urman & Makhortykh, 2023). Potential information quality issues can arise as information is decontextualized and repurposed (Stvilia et al., 2007). Moreover, decades of human-centered research on search engine use have identified consistent patterns that are likely to carry over to the use of GenAI search engines: media and information literacy levels are low (Haider & Sundin, 2019); search engines are trusted and afforded cognitive authority (Huvila, 2013); cognitive biases influence information-based decisions (Azzopardi, 2021); and people are susceptible to inaccurate information (Rubin, 2019). Further, the ability of LLMs to interact in a human-like way and generate text seemingly imbued with meaning and understanding, while possessing no human traits or principles, creates additional vulnerabilities to users of these systems (Bender et al., 2021). In short, the potential is high for unethical and harmful outcomes of the uncritical and unregulated adoption of hybrid Generative AI search engines (GenAI SEs).

In this study, we sought to understand how GenAI responses are designed in relation to topics and queries that represent issues of public concern and welfare. The study was guided by the following research questions. RQ1: Are Generative AI responses influenced by the sentiment of the query and topic? RQ2: How do Generative AI search engines establish authority in their responses to queries? Given the responsiveness of LLM applications, such as ChatGPT, to fine-tuning of style and presentation parameters, we reasoned that an analysis of the responses would reveal the goals of the system designers and inform our understanding of potential impacts on users. In consideration of the potential for system responses to influence users' perspectives, we were particularly interested in *sentiment bias*, reflected in skew in the emotional valence of responses (Huang et al., 2020; Tian et al., 2023), and in tactics used to present claims and establish authority. Findings contribute to a rapidly growing body of work on GenAI search systems, with the goal of increasing public awareness of their strengths and limitations, informing designers

of areas for improvement, and encouraging information researchers and professionals to continue to monitor developments and educate users.

We conducted an audit of public-facing and freely available ChatGPT and Generative Pre-trained Transformer (GPT) implementations in search engines: Perplexity and Microsoft Bing Chat. While Bing Chat has since been renamed as Microsoft CoPilot, we retain the name used at the time of data collection (June 2023). In the sections to follow, we review prior work, outline the methods of data collection and analysis, and present findings in two phases: a quantitative comparison of three systems (ChatGPT version 3.5, Bing Chat, and Perplexity), followed by an inductive analysis of the responses and sources provided by the latter two GenAI search engines. We then discuss the findings and conclude the paper. The present study expands on the initial findings reported by Li and Sinnamon (2023).

PRIOR WORK

The rapid deployment of Generative AI-powered search tools over the past 2 years has been accompanied by major media coverage and a burst of related research activity, much of it only available in pre-print. Terminology is still evolving, with researchers variously describing these systems as LLM-enabled or LLM-based search engines (Kelly et al., 2023; Wazzan et al., 2024); LLM-based chatbots (Urman & Makhortykh, 2023), AI agents (White, 2024), and Generative AI or Generative search engines (Capra & Arguello, 2023; Liu et al., 2023). We adopted the term Generative AI search engine (GenAI SE) to represent these hybrid systems, which combine LLMs with search results to generate responses and link to current Web content.

Evaluation

A number of published papers have evaluated various inputs and features of GenAI SEs. Roegiest and Pinkosova (2024) examined the readability of training datasets of LLMs and GenAI SEs and found that Flesch-Kincaid Grade Levels varied across several datasets from grades 6 to 9, to grades 10-12, which has implications for suitability for different user populations. An exploratory study compared the number of sources provided by the Bing Chat and a traditional search engine for the same search tasks and identified that Bing Chat returned significantly fewer links (4) of which only 2 were cited in the response text, as compared to DuckDuckGo (10 links per page) (Kelly et al., 2023). Another team of researchers audited the verifiability of citations in several GenAI SEs: Bing Chat, NeevaAI, Perplexity.ai, and YouChat (Liu et al., 2023). They found that 52% of the generated sentences were supported by citations, and 75% of responses were fully supported by citations. Perplexity had more citations that supported its sentences than Bing Chat, while Bing Chat more accurately paraphrased from the cited websites compared to Perplexity. An audit study of political bias and censorship compared 3 GenAI systems: ChatGPT, Google Bard, and Bing Chat (Urman & Makhortykh, 2023). The study found a high level of refusal to answer in Google Bard for queries about Putin (90%) compared to other political figures (e.g., Biden), and reported that Bing Chat provided less false information compared to the other systems (Urman & Makhortykh, 2023). A limitation of existing audit studies is a lack of adherence to best practices for algorithm audits (Roegiest & Pinkosova, 2024; Urman & Makhortykh, 2023). Studies that collect data at a single point in time may fail to capture broader patterns and dynamic changes in the algorithms and responses (Metaxa et al., 2021b). Given that LLM systems are not rule-based, but rather employ deep-learning algorithms, outputs may vary substantially over repeated interactions.

Sentiment Bias

Bias in SEs is any type of systematic skew that causes harm (Friedman & Nissenbaum, 1996). Many types and sources of bias influence SEs, including those that arise from system design and user interactions (Baeza-Yates, 2018; Friedman & Nissenbaum, 1996; Kay et al., 2015). Prior research has identified search engine bias based on gender and race (Kay et al., 2015; Metaxa et al., 2021a; Noble, 2018). In this study, we focus on the presentation bias in GenAI system outputs. The tendency for search system output to be influenced by the specific terms, tone and intent of the query has long been understood (Haider & Sundin, 2019; Tripodi, 2018). One manifestation of this is sentiment bias (Huang et al., 2020; Tian et al., 2023). For example, GPT-2 was found to generate text with more positive emotion for the profession “baker” compared to “accountant” (Huang et al., 2020). Sentiment bias can arise from social bias in the training data, interaction bias, or other sources (Baeza-Yates, 2018; Bender et al., 2021; White, 2024). Studies have found that sentiment varies by topic in web search results (Demartini & Siersdorfer, 2010; Kazai et al., 2019; White, 2013). Health-related queries retrieved more confirmatory and positive highly ranked results (White, 2013); whereas controversial queries (e.g., genetic cloning) retrieved results with more negative-oriented emotion (Kazai et al., 2019). People demonstrate preferences (selection bias) for search results with more positive sentiment, such that sentiment bias can impact information behaviour (Kazai et al., 2019; White, 2013). These findings also raise questions about *confirmation bias*, where users prefer search results that echo or reaffirm their preexisting beliefs (White, 2013).

Authority and quality in search results

Search engines are trusted and familiar tools that have acquired the status of *cognitive authorities*: influential sources of expertise (Wilson, 1983; Haider & Sundin, 2019). They function as communicators of knowledge and play a role as cognitive authorities in persuading and influencing public opinion (Hovland et al., 1953). Huvila (2013) found that cognitive authority is afforded both to search systems and to the activity of searching, which may change as systems respond with answers rather than search results. Research has extensively explored how people determine source credibility, with a consistent finding that people often rely upon heuristics (rules, cues, shortcuts) rather than in-depth analysis of features (e.g., Arazy et al., 2017; Hilligoss & Rieh, 2008; Rieh, 2002; Sundin, 2011). Taraborelli (2008) attributes this to a reliance on predictive judgement of quality, based on reputation, rather than evaluative judgement. With respect to GenAI SEs, it is unclear if and how these systems will establish authority, although many users are likely to uncritically accept them as next-generation search engines. Early research suggests that GenAI SE outputs appear to be fluent, coherent (Bender et al., 2023; Zelch et al., 2024), and informative (Capra & Arguello, 2023; Liu et al., 2023; Zelch et al., 2024). Attribution, the capacity of a system to provide evidence and support for claims, for example, through citation of sources, is an important aspect of authority, and, given the nature of LLMs, is one of the challenges currently facing the research community (Li et al., 2023). This study will explore attributions and rhetorical features associated with authority.

User perception of generative AI search engines

A small number of user studies of these new systems have been published, including university students and staff (Capra & Arguello, 2023; Wazzan et al., 2024) and crowdsourced workers (Liu et al., 2023; Malaviya et al., 2023; Zelch et al., 2024). A general observation across these studies is that participants were not aware of how information was generated (Capra & Arguello, 2023; Zelch et al., 2024), such as information about stocks (Capra & Arguello, 2023) or information that includes advertisements (Zelch et al., 2024). Zelch et al. (2024) called attention to *native advertisement* which they used to describe incidences of subtle product and brand placement in the generated text. Students had different levels of trust in the generated responses on informational content (Capra & Arguello, 2023), which may be attributed to the critical thinking skills gained in university. User studies are challenging to conduct, as these systems are evolving so rapidly, and practices associated with such systems are not yet established. While it is likely that established patterns of interaction with SEs will carry over, trust in AI is a subject of intense debate in society, which may influence the uptake and use of these systems at least by some, as suggested by Capra and Arguello's (2023) early study. As this field is new, there are many open questions. Prior research indicates a need to invest research effort in auditing the sentiment of GenAI systems using best practices and investigating the nature of the responses, including the ways in which claims are presented and supported.

RESEARCH METHODS

We designed the study as an *algorithmic audit*: a method that systematically queries an algorithm, observing the responses to examine how the algorithm works (Sandvig et al., 2014). Algorithmic audit has been used extensively in examining general web SEs (e.g., Le et al., 2022), image SEs (Metaxa et al., 2021a), and video SEs (Hussein et al., 2020). The method is derived from traditional auditing practices in the social sciences that examine human discrimination against individuals and groups, but with a focus on automated processes (Metaxa et al., 2021b). We conducted a *scraping audit*, in which repeated queries were used to observe system responses (Sandvig et al., 2014). Algorithmic audits are designed around two axes: the topic and the type of discrepancy (Metaxa et al., 2021b). To probe GenAI systems on bias and authority, we identified four topics representing current global issues: Climate Change, Vaccination, Alternative Energy, and Trust in Media (Edelman, 2023; Government of Canada, 2018). Public knowledge and perspectives on these issues vary substantially among the general population and are significant in shaping policy and behaviour. The type of discrepancy examined in the first stage of the analysis is sentiment bias. We operationalized sentiment bias as a correlation between query sentiment polarity (positive, negative, neutral) and response sentiment polarity. The second stage of our analysis was limited to the responses of the two GenAI SEs (Bing Chat and Perplexity), which, unlike ChatGPT, are designed to function as alternatives to traditional SEs and include citations to sources. By analyzing the responses collected through the audit, we further explored their nature with respect to several additional variables (Table 2).

Systems

We selected three GenAI systems freely available for public use: GPT-3.5 powered ChatGPT and Perplexity (AI team, 2023; OpenAI, 2023b) and GPT-4 powered Bing Chat (Mehdi, 2023). These systems are linked through the shared use of Open AI's GPT model. Early versions of Perplexity also relied on the Bing index, but all three systems have developed rapidly since they were released, and their precise implementations are proprietary and not known. Perplexity and Bing Chat both use *Retrieval Augmented Generation* (RAG), a model that retrieves web information from a search index, and uses that information to generate a system response, including attributes to source references, using an LLM (Lewis et al., 2020; Smith, 2024; White, 2024). At the time of data collection, Google's

LLM-based system (formerly Bard; now Gemini) was not available in the country where this research was conducted (Canada), and therefore, was not included in the study.

Topics and Queries

We use the term “query” to collectively refer to prompts and queries. For each of the four topics, we gathered a set of 12 authentic queries, for a total of 48. The number of queries is consistent with other SE algorithm audits (Hussein et al., 2020; Le et al., 2022). We gathered the queries from AnswerThePublic, a fee-based Web service that scrapes popular searches from Google autosuggestion data (NP Digital, 2023). From among the most frequent English-language queries originating in Canada in May 2023, we selected a set of queries with substantial sentiment variation within each topic (Table 1). Sentiment was calculated using a fine-tuned transformer model (Yuan, 2023), ranging from -1(negative sentiment), 0 (neutral sentiment), to 1(positive sentiment) polarity scores. The queries were submitted in their basic form, with no additional tuning prompts, in order to retrieve responses based on the default settings.

Topic	Query	Sentiment Polarity
Climate Change	will climate change kill us	-0.57
	will climate change ever stop	-0.06
	climate change benefit	0.76
Vaccination	why vaccine mandates are unethical	-0.61
	vaccination in pregnancy	-0.08
	vaccination benefits	0.58
Alternative Energy	why alternative energy is bad	-0.75
	alternative energy advantages and disadvantages	0.04
	renewable energy is important	0.84
Trust in Media	distrust in media	-0.72
	trust in media all time low	-0.10
	trust in media by country	0.37

Table 1. Example Queries and Sentiment Polarity Scores (-1 to +1).

Data Collection

Best practices for algorithmic audits include conforming to legal and ethical guidelines regarding data scraping (Metaxa et al., 2021b). While we found no guidelines for Bing Chat (Bing, 2023), Perplexity and ChatGPT disallow automatic data scraping (OpenAI, 2023a; Perplexity, 2023). Therefore, we collected the data through manual query submission and documentation of responses and sources. For seven days during one week in June 2023, 12 different search queries for each topic (a total of 48 queries) were submitted daily to each of the three systems (Bing Chat, Perplexity, and ChatGPT). A total of 1,008 responses were collected in the study: 7 days x 48 queries (across 4 topics) x 3 systems. The rationale for submitting the same 48 queries to each system over multiple days was to anticipate dynamic changes in the algorithms and responses, as the goal of this audit was to examine broader trends, and not to target specific phenomena at a specific time (Metaxa et al., 2021b). Several different dummy accounts were used for ChatGPT that required sign-in, and data was collected via different public computers on a university campus to avoid the effects of personalisation. Private mode was used to access Bing and Perplexity anonymously via Microsoft Edge browser. Default settings were used for each system to mimic non-expert information seeking behaviour. Although these systems invite conversational style interactions, we issued a single query and recorded the first response without iterations. A fresh session was initiated after each query to reduce impacts of session memory.

Data Preparation and Analysis

There were two phases of this study: automated analysis, followed by response and source analyses. Table 2 presents the variables and measures. In the first phase of the analysis, Python (version 3.10) in the Spyder integrated development environment was used to pre-process the collected data ($N = 1,008$) and to calculate response length and sentiment polarity scores of queries and responses for the three GenAI systems (ChatGPT, Bing Chat, Perplexity). Data cleaning was carried out using Regular expression (version 3.10.12; Python Software Foundation, 2023), Pandas (version 1.5.3; McKinney, 2010), and NumPy (version 1.25.0; Harris et al., 2020) packages. In most cases, system responses to the same queries differed across the 7 days, but a small number of duplicate responses by Bing Chat were retained in the analysis, as they were collected on different days, computers, and accounts. Statistical analyses of sentiment polarity were conducted using IBM SPSS (version 29.0.1) to assess correlation and group differences by topic and search system.

The second phase focused on the responses of GenAI SEs: Bing Chat and Perplexity ($N = 672$). The readability of responses was assessed using the Flesch Kincaid Grade Level metric (Kincaid et al., 1975), calculated using the

Textstat package (version 0.7.3, Bansal & Aggarwal, 2022) in Python. We then manually analyzed responses following a general inductive coding approach as outlined by Thomas (2006). The primary objective of the inductive analysis was to identify rhetorical tactics used to support claims in relation to the query and to establish authority. The two authors independently carried out open coding on a purposive sample of 30 responses that varied by sentiment polarity, topic, and system. We then met to compare outcomes and jointly devised a coding framework around four identified parameters: narrative voice of responses, stance in relation to query, and techniques to support and to mitigate claims. Codes and definitions are presented together with the results in Table 4. The unit of analysis was set at the response level (i.e., a code is either present or not present in a response). The first author coded all responses and a coder who was not involved in the project coded a 10% random sample of responses ($n = 67$) (O'Connor & Joffe, 2020). Following an initial low inter-coder reliability score ($\alpha = 0.59$) (Krippendorff, 2018), the coders met, identified the main area of disagreement, and revised the codebook for greater clarity. A subsequent coding round resulted in a reliability score of $\alpha = 0.71$, which is an acceptable strength of agreement (Krippendorff, 2019). Calculations of reliability scores were carried out in Python using the Fast Krippendorff package (version 0.6.1.; Castro, 2023). Results of the readability and rhetorical analyses are summarized using descriptive statistics and elaborated in a narrative reporting of results.

Variable	Measure	Source of Measure
<i>Phase 1: ChatGPT 3.5, Bing Chat and Perplexity</i>		
Query sentiment	Polarity score (-1 to +1)	Python 3.10, a transformer model
Response sentiment	Polarity score (-1 to +1)	Python 3.10, a transformer model
Length of response	Word count	Python 3.10
<i>Phase 2: Bing Chat and Perplexity</i>		
Readability of response	Flesch Kincaid Grade Level	Python 3.10, Textstat package
Rhetorical features	4 feature types and values (details in Table 4)	Inductive coding, manual
Number of citations	Source count	MS Excel
Source domain type	14 Source type categories	Manual classification
Source origin	Country or Region	Manual classification

Table 2. Study Variables, Measures, and Sources of Measures

The sources cited in responses ($N = 3448$) were captured and analyzed at the level of the web domain (e.g., www.cdc.gov), rather than the specific page level. We manually reviewed the 355 unique web domains to identify the geographic origin and to classify the source by type of media or organization. Using an inductive and iterative process, we identified 14 categories of sources from the data, for example, News and Media, Commercial Business, and Government, and assigned each domain to a single category. The full category list is included in Table 5. We then aggregated domains by system and by topic to conduct a quantitative descriptive analysis.

RESULTS

A descriptive summary of the 1008 responses by system and topic is presented in Table 3. The average length of results across all topics and GenAI systems is 234 words. Within each system, response length does not vary substantially within each topic, but there is a notable difference across systems. ChatGPT responses are the longest ($M = 336$), followed by Perplexity ($M = 243$) and Bing Chat ($M = 123$). The number of cited sources for Bing Chat and Perplexity responses is remarkably consistent across systems and topics at just over 5 sources per response, whereas ChatGPT does not cite sources. The following subsections present findings from the first phase (sentiment bias) and the second phase (response and source analysis).

Topic	ChatGPT*		Bing Chat		Sources	Perplexity		
	Length <i>M(SD)</i>	Length min/max	Length <i>M(SD)</i>	Length min/max		Length <i>M(SD)</i>	Length min/max	Sources <i>M</i>
Climate Change	347 (85)	189/500	139(51)	56/291	5.19	283(96)	156/597	4.96
Vaccination	297(115)	67/493	117(42)	35/224	5.10	225(71)	88/428	5.25
Alternative Energy	346(90)	114/528	117(35)	49/227	5.02	214(50)	90/426	5.27
Trust in Media	353(82)	91/635	121(34)	62/260	5.19	250(58)	149/505	5.06
Total – all topics	336(96)	67/635	123(42)	35/291	5.13	243(78)	88/597	5.14

Table 3. Response Length (words) and Number of Sources per Response by System and Topic.

*No sources cited

Sentiment Bias

Query Sentiment Bias

Preliminary analyses ($N = 1,008$) show the relationship between query polarity and GenAI system response polarity is linear with both variables normally distributed, as assessed by Shapiro-Wilk's test ($p > .05$), and there are no outliers. Thus, a parametric Pearson's product-moment correlation is utilized to assess the relationship. The analysis identifies a moderate positive correlation between query polarity and response polarity, $(1,006) = .46, p < .001$, with query polarity explaining 21% of the variation in response polarity. Similar levels of correlation are found for all three systems: ChatGPT ($r = .435, p < .001$), Bing Chat ($r = .454, p < .001$), and Perplexity ($r = .509, p < .001$). In other words, a more positive query leads to a more positive response in all three GenAI systems.

System and Topical Sentiment Bias

Shapiro-Wilk's tests show non-normal distributions of response polarity for all three systems and topics ($p < .001$), so we use non-parametric Kruskal-Wallis H tests to test for differences. Median response polarities are significantly different between groups for the three systems, $\chi^2(2) = 12.54, p = .002$, and the four topics, $\chi^2(3) = 299.36, p < .001$. Pairwise comparisons are conducted using Dunn's procedure, with a Bonferroni correction. Adjusted p-values are presented. Perplexity has significantly lower polarity of responses ($Mdn = -.19$) than Bing Chat ($Mdn = -.10, p = .05$) and ChatGPT ($Mdn = -.01, p = .05$). Trust in Media ($Mdn = -.39$) and Climate Change ($Mdn = -.33$) have significantly lower polarity than Vaccination ($Mdn = .25, p = .001$) and Alternative Energy ($Mdn = .24, p = .001$). Therefore, Perplexity presents more negative responses compared to Bing Chat and ChatGPT. Responses to Trust in Media and Climate Change topics are more negative than those for Vaccination and Alternative Energy.

Response Analysis

We conducted a deeper analysis and comparison of the responses provided by the two GenAI SEs, Bing Chat and Perplexity. The mean Flesch Kincaid Grade Level is higher for Perplexity ($M = 12.82; SD = 2.00$) than Bing Chat ($M = 9.87; SD = 2.11$), indicating that Perplexity responses are designed for a higher educational reading level (post-secondary) compared to Bing Chat (secondary, high school). Beyond readability, we conducted a rhetorical analysis of responses, which is summarized in Table 4. For categories 1 and 2, responses were assigned a single code or no code if none applied (mutually exclusive). In categories 3 and 4, all instances of tactics were coded in each response (non-mutually exclusive).

Findings for narrative voice (1) indicate that Bing Chat tends to respond in the first person, for example, often adding, "I hope this helps." Perplexity often uses the third person and passive voice, with most responses referring to "search results" or to content from search results as the authority upon which the response is based.

In relation to the query (2), most responses in both systems support explicit or implicit claims in the query. For example, in response to, *will renewable energy stocks go up*, Bing Chat states, "Renewable energy stocks have been growing significantly for the last decade and are anticipated to continue this trend." Perplexity's response to *vaccination pros and cons* is supportive, and more controversial: "Experts who are against immunization disagree and actually find vaccines to be harmful to health because they cause adverse side effects." Cases of query opposition are limited (10-12%), and primarily occur in response to counter-factual queries, such as, *climate change is not caused by humans and why alternative energy is bad*. The two systems take different approaches when disagreeing. For example, some Bing Chat responses to *why vaccine mandates are unethical* note that it is a controversial topic with different opinions, while other responses clearly state that mandates are ethical, citing health authorities. Perplexity responses to this query simply identified it as a controversial topic.

Feature	Definition and example	Bing Chat (n = 336)	Perplexity (n = 336)
1) Narrative voice of response (mutually exclusive)			
First person singular	Makes claims as an individual author - “I”	81.0%	3.9%
First person plural	Makes claims as group author - “we”	5.4%	10.1%
Third person	Attributes claims to other authorities - “search results”, passive voice	13.6%	86.0%
2) Stance in relation to Query (mutually exclusive)			
Support	Agrees with/support viewpoint expressed in the query	81.3%	81.0%
Neutral	Presents a balanced viewpoint in relation to the query	6.8%	8.9%
Oppose	Disagrees with/opposes viewpoint expressed in the query	11.9%	10.1%
3) Techniques used to support claims (non-mutually exclusive)			
Generic authorities	References general authorities – e.g., scientists, most people	56.2%	56.0%
Named authorities	Referenced specific named entities – e.g., NASA	53.3%	53%
Quantification	Numerical data and/or statistics are referenced	51.2%	45.2%
4) Techniques used to mitigate claims (non-mutually exclusive)			
Balance check	Offers a counterpoint or alternative perspective - “Some people believe... However, others believe...”	25.9%	39.0%
Conditional language	Notes that situations differ based on time, location, policy - “Different factors ... contexts can influence...”	14.6%	16.7%
Hedging	Uses language to add uncertainty (e.g., can, could, might, may, potential) and reduce specificity (e.g., some, many)	74.1%	83.6%
Notes limitations	Mentions shortcomings of its own knowledge – “As an LLM, I cannot...”	0.3%	2.4%

Table 4. Summary of Rhetorical Features and Frequencies in Responses of Bing Chat and Perplexity

To support their claims (3), more than half of the responses in Bing Chat and Perplexity refer to generic authorities, (often researchers or scientists) and/or named authorities. Perplexity frequently referenced vague, majority, or minority viewpoints, such as “most people” or “some people.” In response to the query *vaccination pros and cons*, Perplexity offers both perspectives: the pro arguments tend to start with assertions such as “Vaccines can save...” and “Vaccines help...”, the con arguments start with “Some people...”. It is unclear if such subtle techniques of differentiation are picked up in the training data or designed into the response generation process. In some cases, named authorities are companies or products. For instance, Bing Chat recommended stocks (e.g., Canadian Solar, Vestas Wind Systems, NextEra Energy) in response to the query *will renewable energy stocks go up*; and Perplexity frequently mentioned a specific company (Alternative Energy Solutions, Ltd.) when asked about *alternative energy solutions*.

Techniques to mitigate claims (4) are used extensively by both systems, with hedging being the most common. For instance, “Based on the search results, it is difficult to predict with certainty whether renewable energy stocks will go up. However, there are several factors that could influence the performance of these stocks.” Notably, we found very few instances in which the systems admitted the limits or shortcomings of their responses in relation to these topics.

Source Analysis

Responses to the 672 (48 unique) queries submitted to the two systems cited 3,448 sources in 355 unique web domains. Of these domains, 28% were cited only by Bing Chat; 46% were cited only by Perplexity, and 26% were cited by both. 25% were in the long tail: cited only once across the whole dataset, even though each query was submitted 7 times to each system. The average number of citations per response was equivalent for the two systems at 5.1 (5.125 Bing, 5.137 Perplexity). While sources ranged from 1 to 7 for the two systems, the majority had between 4 to 6. Sources originated from 24 countries, mostly in North America and Europe, but including Japan, Pakistan, Hong Kong, South Africa, and Australia. 95% of source domains originated in the United States (U.S.), United Kingdom (U.K.), Canada, and international organizations (Figure 1). 65% of references were to sources in the U.S. Bing referenced more sources from the U.K. and Canada, and fewer from the U.S. than Perplexity.

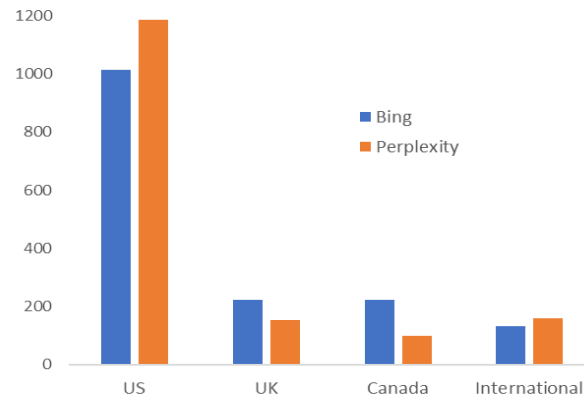


Figure 1. Number of sources from the most frequently cited regions (N= 3190; 95% of total source domains)

Table 5 shows the distribution of source types by web domain, manually classified into 14 categories. Source types are listed in decreasing order of frequency, and those greater than 10% are bolded. The most frequently cited category is News and Media (21.8%), which includes mainstream news and broadcasting organizations, followed by Business (17.9%), Government (16.5%), Non-Profit (13.3%) and Digital Media Site (11.7%). *Digital media sites* are informational websites focused on specific domains and they range widely in quality. Nine of the 14 categories are cited infrequently, including several typically associated with high quality information: Educational Institutions (2.6%), Academic Publishers and Journals (3.0%), Policy Institutes (1.7%) and Professional and Scholarly Associations (3.1%).

Source Types	Bing Chat					Perplexity					Total
	CC	Eng	Med	Vacc	Total	CC	Eng	Med	Vacc	Total	
News and Media	39.9	11.0	35.3	18.3	26.2	22.6	10.0	29.6	6.7	17.4	21.8
Business	0.5	40.0	15.4	2.3	14.3	4.6	43.6	25.7	10.7	21.6	17.9
Government	25.0	2.4	3.3	36.2	17.0	32.2	7.0	0.0	26.9	16.1	16.5
Non-Profit	8.3	2.6	23.0	6.7	10.1	17.5	11.1	18.1	19.5	16.5	13.3
Digital Media Site	11.5	21.1	7.1	20.6	15.1	6.3	5.5	3.6	18.7	8.3	11.7
NGO	6.9	14.1	0.0	6.4	6.8	4.8	11.1	0.0	4.0	5.0	5.9
Association	2.5	1.2	0.0	2.1	1.5	8.4	4.1	1.6	5.0	4.7	3.1
Acad. Publisher	0.0	1.7	9.5	0.9	3.0	0.5	0.2	6.1	5.2	3.0	3.0
Educational Inst.	2.1	4.3	0.0	3.9	2.6	0.7	4.1	3.4	2.0	2.6	2.6
Policy Institute	0.0	0.2	4.5	0.0	1.2	0.2	1.4	7.0	0.0	2.2	1.7
Social Media Site	1.6	0.7	1.9	0.7	1.2	1.9	0.2	4.3	0.0	1.6	1.4
Personal Website	0.9	0.0	0.0	1.6	0.6	0.0	0.0	0.0	0.0	0.0	0.3
Search Engine	0.0	0.7	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1
Other	0.9	0.0	0.0	0.2	0.3	0.2	1.6	0.7	1.2	0.9	0.6
Grand Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Table 5. Distributions of Source Types (%) by System and Topic (CC=Climate Change, Eng=Alternative Energy, Med = Trust in Media, Vacc=Vaccination); >10% in bold

Examining the sources by system reveals some differences. Bing Chat cites a greater proportion of sources from News and Media, and Digital Media domains than Perplexity, which cites relatively more Business and Non-Profit source domains. There are clear patterns in source citation across topics. In both systems, the Climate Change and Vaccination topics cite Government sources heavily; Business sources are most frequently cited for the Alternative Energy topic; and News and Media and Non-Profit sources are most frequently cited for the Trust in Media topic.

With respect to information quality, several questions arise from this analysis, including the frequent citation of Digital Media Sites by Bing Chat, particularly for the Vaccination topic (20.6%), and the prevalence of Business sources, particularly for the Alternative Energy topic (>40%). In the first case, sources include popular consumer and medical health websites, including WebMD, Healthline, and VeryWell Health, as well as some less authoritative

sites, such as Grunge, rated as “questionable” by Media Bias/ Fact Check (2024). In the second case, for the Alternative Energy topic, Perplexity cites approximately 30 different company websites (e.g., Inspire Clean Energy) that offer a range of energy services, many of which include blog posts and articles about renewable energy, as well as finance, investment, and employment sites related to the energy sector. Across these 355 domains, we identified other questionable and poor-quality sources, including essay writing services (e.g., IvyPanda), a high school biology assignment, climate change denial articles, and advertisement-based content farms. While the most frequently referenced domains are relevant and authoritative, including government and non-governmental agencies (e.g., the U.S. Center for Disease Control, Government of Canada, World Health Organization), research organizations (e.g., Gallup Consulting and Pew Internet Research), and major news agencies (e.g., British Broadcasting Corporation), this level of quality is not consistent.

DISCUSSION

In response to RQ1, we found a significant impact of both query bias and topical bias in the GenAI system responses. A medium-strength correlation was found between the query and responses suggesting that responses echoed the emotional valence of the query. While cause and effect cannot be inferred, we varied the query polarity within each topic, so topic or domain effects are not likely to be factors. Results with respect to query stance, showing that most responses supported or were neutral in relation to the query, lend further support to the conclusion that responses are influenced by the wording and tone of the query, as has been observed in studies of traditional SEs (Tripodi, 2018; White, 2013). Query sentiment and stance bias may serve to limit the diversity of perspectives encountered by searchers and reinforce confirmation bias (Azzopardi, 2021). Topical bias was reflected in more positive sentiment for solution-focused topics (Vaccination and Alternative Energy) over problem-focused topics (Climate Change and Trust in Media). This is not unexpected and likely reflects the sentiment in the model and the underlying web content, in keeping with the results of prior research (Demartini & Siersdorfer, 2010; White, 2013).

In response to RQ2, we presented various indicators of authority and quality in GenAI SE responses. An interesting difference arises from the choice of first-person narration by Bing Chat and Perplexity’s use of third person and passive voice. This reflects a different approach to establishing cognitive authority, which can be situated in different actors, elements or even activities (Hovland et al, 1953; Huvila, 2013). Bing Chat adopts the role of authoritative knower and “communicator,” while Perplexity presents as an intermediary, passing on knowledge from search results and authoritative others. The second approach more faithfully represents the hybrid, RAG, model used by GenAI SEs, in which the LLM is prompted with search results and constructs a response from them. It remains to be seen if these different rhetorical approaches will shape users’ mental models and impact perceptions of trust and credibility.

These responses are clearly designed to serve as authoritative, evidence-based sources of information for these topics. Both systems use a range of common techniques to support claims, including reference to specific and general authorities and statistics. At the same time, they make heavy use of rhetorical hedging and equivocation, which reduces the strength of arguments, and perhaps more importantly for these systems, limits the responsibility and reputational risk of the information providers. Both systems employ balance checks and surface minority voices (“some people”) to present different perspectives, even for topics with clear factual grounding in science. Balance checks are associated with pseudo-science and the problematic media legacy of prioritizing apparent neutrality and balance over scientific consensus in issues such as vaccination and climate change (Boykoff & Boykoff, 2004), so it is disturbing to see such techniques on display. As noted by Ghezzi and colleagues (2020), it is unethical for SEs to provide anything but the highest quality, evidence-based information on critical issues such as vaccination. Such techniques pose the danger that bad actors may take advantage of these outputs as LLMs are not subject to truth or ethics (Bender et al., 2021).

Our examination of sources found levels of attribution similar to Kelly and colleagues (2023). Unlike some prior work (Liu et al., 2023; Malaviya et al., 2023; Shi et al., 2023), we did not assess the validity of the specific sources or the extent to which they support claims made. Nevertheless, our domain-level analysis offers valuable insights. Notable is the heavy dependence on news and media content across systems and topics. This highlights the complex relationship between media organizations and this new generation of search engines, which no longer function as directories, but consume media content to produce an information service. More surprising is the prominence of business websites, particularly in Perplexity responses. This evident commercial bias may be an outcome of search engine optimization and the practice of publishing blogs and articles on company websites to increase ranking and traffic. Equally, this may be an intentional step towards a form of native advertising (Zelch et al., 2024) and a potential revenue model. Either way, both systems seemed to interpret the renewable energy topic as commercial in intent, which was not our interpretation in designing the study. In addition to commercial bias, we found evidence of geographic bias in the heavy reliance on sources from the U.S. and other Western countries. On a practical level, the low number of sources from Canada is surprising, given the queries were issued in Canada. However, the extent to

which these systems employ geographic personalisation is not known. Bing Chat did provide more results for the vaccination query from Canada, which suggests some use of geolocation data. More broadly, it will be important to determine the extent to which regional and linguistic bias places limits on these systems.

With respect to quality, while many of the cited sources cited are appropriate and authoritative, the value of these systems is seriously undermined by the seemingly random inclusion of poor-quality sources of information, including commercial blogs, clickbait, essay mills, and content farms. Yes, the Internet is full of such content, but given the nature of the topics in this study, such sources are wholly inappropriate. If they were retrieved as one item in a ranked list, they could be easily assessed and passed over, but with GenAI SEs, each source is integral to the response, so the quality of each resource must be held to a higher standard.

Finally, we found unmistakable evidence of design differences between these systems. Bing Chat responses are shorter, more conversational, easier to read, more positive in sentiment, and results are drawn from a narrower set of domains in comparison to Perplexity. Perplexity uses a more academic tone and approach, and the reading level is set higher than other available chatbots (Roegiest & Pinkosova, 2024). At the same time, it is more U.S.-centric and commercial in orientation, at least for these topics. The subtle differentiation of these tools points towards a path in which search engines will become even more adaptive and hyper-personalised.

CONCLUSION

This study makes several contributions to this rapidly developing area of research on GenAI SEs. Following best practices for algorithmic system audits, we found evidence of sentiment bias in relation to queries and topics, and commercial and geographic biases in the cited sources. The different profiles adopted by Bing Chat and Perplexity highlight the extent to which GenAI SEs can be tailored and potentially influential in shaping public opinion in different directions. Responses clearly seek to establish cognitive authority by making claims and providing evidence, and yet the strength of these claims is undermined by ambiguous language and sources of questionable quality. We acknowledge the limitations of this study and the need for further work. We focused on 4 broad topics of public importance at one point in time (June 2023), and future work will need to examine a wider range of topics and tasks to claim evidence of general patterns. The orientation of the study was to consider the human impacts of these systems, in keeping with the conference theme of “people first,” but our method did not involve human participants. In our future work, we will design user studies to test and validate these findings in relation to authority, credibility, and source quality.

GENERATIVE AI USE

The outputs of Generative AI systems were the subject of this study. We did not use them in any other steps of analysis or writing. The authors assume all responsibility for the content of this submission.

AUTHOR CONTRIBUTIONS

Li & Sinnamon: Conceptualization, formal analysis, funding acquisition, methodology, project administration, writing. Li, Sinnamon, & Hsueh: Investigation. Li & Hsueh: Software. Li & Li: Validation. (Contribution roles according to Contributor Roles Taxonomy (CRediT)- <https://credit.niso.org/>)

ACKNOWLEDGMENTS

We thank Shawn Ching-Chung Hsueh (hsueh@phas.ubc.ca) and Jason Li (jasonli5@andrew.cmu.edu) for their support in the data collection and analysis stages, and anonymous reviewers for their suggestions. This work is supported in part by funding from the Social Sciences and Humanities Research Council of Canada (SSHRC) CGS-D program, the Natural Sciences and Engineering Research Council of Canada (NSERC) CREATE program, and the Anne and George Piternick Student Research Award from the School of Information, University of British Columbia.

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